

CyberFuse-CI: Adversarially Resilient Vulnerability Detection Through Heterogeneous Multi-Source Data Fusion and LLM-Augmented Reasoning

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Abstract - Enterprise vulnerability detection remains a fundamental challenge in cybersecurity. Organizations face fragmented data streams, including CVE repositories, STIX threat feeds, SIEM logs, and code repositories, that don't talk to each other. Each uses different schemas, and there's no unified way to correlate them. We built CyberFuse-CI, a three-layer Computational Intelligence framework, to address this gap. It normalizes all four source types to a common schema, links entities through an LLM-augmented knowledge graph built with BiLSTM-CRF named entity recognition, Graph Attention Network link prediction, and GPT-4 candidate link validation, then classifies each entity using an adversarially hardened XGBoost ensemble trained with Fast Gradient Sign Method perturbations at $\epsilon = 0.01$. Every detection maps to a MITRE ATT&CK v15 technique with a corresponding remediation control. On the CVEfixes benchmark and a 90-day anonymized enterprise SIEM corpus, CyberFuse-CI reached 94.7% F1-score under normal conditions and held 91.2% accuracy under FGSM adversarial perturbation. We're identifying zero-day candidates a mean of 6.2 days before NVD publishes the corresponding CVE. That's a window that simply doesn't exist with single-source tools. Code and evaluation scripts are in the companion repository [7].

Keywords: adversarial machine learning; computational intelligence; heterogeneous data fusion; LLM-augmented reasoning; vulnerability detection; MITRE ATT&CK; cybersecurity.

I. INTRODUCTION

Vulnerability discovery has exploded in recent years. The CVE program alone recorded over 40,000 vulnerabilities in 2024, and we've already seen more than 12,000 new entries in just the first quarter of 2025 [1]. Manual review simply can't keep up anymore, and even our automated detection systems are buckling under the pressure.

Current detection tools suffer from tunnel vision. Each one looks at a single data stream in isolation. Static scanners examine source code. Intrusion detection systems watch network traffic. Threat intelligence platforms consume STIX feeds. The problem? Each tool sees only part of the picture. In our experience, zero-day exploits rarely announce themselves in a single data source. They show up first as a STIX threat actor indicator, then as a SIEM anomaly, and only later as a published CVE. A single-source tool misses at least two of those three signals every time.

Large Language Models have opened new directions in vulnerability analysis. A comprehensive systematic review covering over 300 LLM applications in cybersecurity identifies vulnerability detection, program repair, and threat intelligence extraction as the highest-activity application domains, with

fine-tuned models demonstrating strong benchmark performance [2]. A dedicated survey of 263 vulnerability detection studies finds decoder-only models such as GPT-4 and CodeLlama now account for 65% of fine-tuning experiments, driven by stronger cross-language generalization [3]. LLMs used as cybersecurity reasoning engines do carry documented failure modes though: over-attribution based on superficial feature co-occurrence, unstable outputs when source data conflicts, and poor generalization on attack surfaces outside the training distribution [4]. We ran into all three in early prototypes.

The adversarial robustness problem compounds this. NIST AI 100-2e2025 catalogs the full attack taxonomy for ML systems in security contexts, from data poisoning at training time to evasion attacks at inference [5]. Intrusion detection models trained only on clean data can lose more than 20 percentage points of accuracy under FGSM perturbation [6]. A detection tool that collapses under targeted adversarial input isn't just unreliable, it's itself a vulnerability.

We built CyberFuse-CI to address both the coverage gap and the adversarial robustness gap within a single pipeline. The framework normalizes CVE and NVD records, STIX 2.1 threat feeds, SIEM event logs, and open-source code repository data

into a unified entity schema. An LLM-augmented knowledge graph then fuses those normalized entities and surfaces cross-source co-occurrence signals that tend to indicate active exploitation before any CVE is formally assigned. An adversarially hardened XGBoost classifier trained with FGSM perturbations handles multi-class severity assignment. Each detection maps to a MITRE ATT&CK v15 technique with a recommended mitigation. All source code and evaluation artifacts are in the companion repository [7].

II. RELATED WORK

The vulnerability detection landscape has shifted dramatically. Researchers have moved away from encoder-only models like CodeBERT and embraced large decoder-only architectures such as GPT-4 and CodeLlama. Zhang et al. [2] reviewed over 300 LLM cybersecurity applications and found vulnerability detection among the most active areas. A recent survey of 263 studies found these newer models now dominate the field, representing 65% of all fine-tuning work [3]. The reason is straightforward: they generalize much better across programming languages than their predecessors. What's still missing, though, is adversarial robustness testing and repository-level benchmarks that reflect real codebases.

For multi-source data fusion specifically, Li et al. [8] fused network telemetry, geopolitical event records, and bilateral trade statistics through a dynamic multi-view GNN with temporal interaction attention and got strong results on state-sponsored cyberattack prediction. The main finding from that work is that different data modalities carry different signals at different times, and fusing them temporally beats any individual modality. That same logic applies to vulnerability data, where CVE, STIX, SIEM, and code repository streams each contribute something the others don't have.

Yin et al. [9] built a vulnerability knowledge graph from CVE records and unstructured threat intelligence text using BiLSTM-CRF extraction, GAT link prediction, and GPT-4 validation, reaching 90.1% F1-score on cybersecurity entity extraction. That is the closest existing work to our Layer 2 design. What it does not cover is adversarial robustness of the classifier that consumes the graph output, which is where CyberFuse-CI extends the approach.

On adversarial robustness for security classifiers, Ghajari et al. [6] mixed clean and FGSM-perturbed NF-ToN-IoT-v2 samples during XGBoost training and brought the clean-to-adversarial accuracy gap below one percentage point. We applied the same training discipline to vulnerability severity classification, operating on 47-dimensional tabular features from a knowledge graph rather than raw network packet features. The technique works. We've verified it here at a different level of the security stack.

Sarker et al. [10] reviewed AI-driven cybersecurity methods broadly and named adversarial resilience and multi-modal data integration as the two leading unsolved problems for enterprise deployment. Both show up as gaps in every prior work we reviewed. CyberFuse-CI was designed with both in mind from the start.

III. THE CYBERFUSE-CI FRAMEWORK

A. Architecture Overview

CyberFuse-CI runs vulnerability data through three sequential steps. The first step ingests and normalizes all four source types to a shared schema. The second step builds a knowledge graph that links entities across sources and surfaces co-occurrence signals. The third step runs the adversarially hardened severity classifier and produces MITRE ATT&CK mappings. Table I shows the components and data flows at each step.

TABLE I: CyberFuse-CI Three-Layer Architecture

Layer	Component	Data Sources and Outputs
Layer 1	Ingestion and Normalization	CVE/NVD API, STIX 2.1/TAXII 2.1, SIEM Syslog/CEF, CVEfixes repository
Layer 2	LLM Knowledge Graph	BiLSTM-CRF NER, GAT link prediction, GPT-4 validation, OWL ontology graph
Layer 3	Classification and Mapping	XGBoost ensemble, FGSM adversarial training, ATT&CK v15 technique mapper

B. Layer 1: Multi-Source Ingestion and Normalization

Every data source in enterprise security uses a different schema. CVE records follow CVSS scoring and CWE classification. STIX 2.1 bundles organize indicators, threat actors, and campaigns as JSON objects linked by typed relationships. SIEM logs arrive in syslog or Common Event Format with free-text fields interleaved with structured headers. Source code repositories expose syntax trees but carry no severity metadata.

Layer 1 resolves this by normalizing all incoming records to a shared intermediate representation: a VulnerabilityEntity tuple containing a unique identifier, a CVSS-normalized severity score on a 0 to 10 scale, the affected component name, version range, and a free-text description. For NVD records, the CVSS v3.1 base score maps directly. For STIX threat intelligence objects, the confidence field (range 0 to 100) is divided by 10 to produce a compatible score. For SIEM events, a rule-based severity classifier assigns a score using event taxonomy and keyword signals from a domain dictionary of over 150 cybersecurity terms. For CVEfixes code commits, the CVSS score from the linked CVE record is inherited. It's a straightforward normalization step, but it's what makes cross-source comparison possible downstream.

C. Layer 2: LLM-Augmented Knowledge Graph Fusion

With four sources normalized to the same schema, the next problem is linking entities that describe the same underlying threat. A BiLSTM-CRF named entity recognizer, augmented with the cybersecurity domain dictionary, runs over each entity description and extracts CVE identifiers, product names, vendor names, and CWE categories. An attention-based CNN then classifies typed relationships between entity pairs, assigning labels from: 'affects', 'has-weakness', 'exploited-by', and 'mitigated-by'. Those typed triples populate an OWL knowledge graph aligned with the MITRE ATT&CK data model.

Graphs built from NER and relation extraction alone tend to have gaps, particularly for implicit links that span source boundaries. We fill those with two mechanisms. TransH-based graph embeddings combined with BERT textual node vectors feed a Graph Attention Network that learns to surface structurally implied links from the graph topology. GPT-4 separately generates candidate link justifications for entity pairs, and we validate each by computing BERT cosine similarity between the justification and the mean embedding of both candidate nodes. Candidates scoring above 0.60 get added to the graph.

Why does cross-source fusion work so well here? Because zero-days rarely announce themselves in a single stream. The knowledge graph surfaces a cross-source co-occurrence query that identifies VulnerabilityEntity nodes sharing the same affected component across at least two distinct data sources. A CVE entity, a STIX threat actor indicator, and a SIEM anomaly event that all reference the same product and version represent a high-confidence active exploitation signal. That three-source co-occurrence count becomes a high-weight feature in the Layer 3 classifier.

D. Layer 3: Adversarially Hardened Classification and MITRE Mapping

Each VulnerabilityEntity coming out of Layer 2 needs three things assigned: a severity tier from the five-class set (Critical, High, Medium, Low, or Informational), a predicted exploit likelihood score, and a primary MITRE ATT&CK v15 technique identifier.

An XGBoost ensemble classifier receives a 47-dimensional feature vector. The features span CVSS sub-scores, CWE category embeddings (10 dimensions, PCA-reduced), graph centrality measures of the vulnerability node (degree, betweenness, closeness, PageRank, clustering coefficient), STIX indicator temporal recency, cross-source co-occurrence counts, NER entity confidence scores, LLM link prediction confidence scores, data source origin flags, temporal recency indicators, and metadata features. We chose XGBoost because it stays robust under adversarial attack and handles imbalanced class distributions well. Neural networks proved too fragile for this task in our early experiments.

Adversarial hardening follows the FGSM procedure from Goodfellow et al. [12]. The perturbation is computed as:

$$\mathbf{x}_{adv} = \mathbf{x} + \epsilon \cdot \text{sign}(\nabla_{\mathbf{x}} L(\theta, \mathbf{x}, y)) \quad (1)$$

where $\mathbf{x}$ represents the input feature vector, ϵ controls perturbation magnitude (we use $\epsilon = 0.01$), $L(\theta, \mathbf{x}, y)$ is the cross-entropy loss function, and θ denotes the classifier parameters. The gradient $\nabla_{\mathbf{x}} L$ tells us which direction maximizes classification error; FGSM perturbs each continuous feature by ϵ in that direction. A preliminary XGBoost model provides gradient sign estimates through finite-difference approximation. We shift each continuous feature by $\epsilon = 0.01$ in the gradient sign direction, leaving categorical and binary flags unchanged. The final model trains on 50% clean and 50% adversarially perturbed samples. We tested epsilon values of 0.005, 0.01, 0.02, and 0.05. Below 0.005 the robustness improvement was negligible; above 0.02 clean accuracy started dropping noticeably. The 0.01 value gave the best tradeoff.

MITRE ATT&CK and MITRE ATLAS mappings use a FAISS IndexFlatIP built from ATT&CK v15 technique descriptions encoded with the all-MiniLM-L6-v2 sentence transformer. At inference, the entity description is encoded and we retrieve the nearest technique by cosine similarity. Each entity gets a technique ID, a confidence score, and a linked mitigation recommendation from the ATT&CK catalog, giving the downstream ticketing system something actionable rather than just a severity number.

IV. EXPERIMENTAL SETUP

We evaluated CyberFuse-CI on two datasets. The CVEfixes benchmark [11] contains function-level commits from open-source repositories that fix CVEs across C, C++, Python, Java, and JavaScript. We used 10,077 vulnerable function samples and 18,406 non-vulnerable samples. The class imbalance is fairly pronounced in CVEfixes, so we applied SMOTE oversampling within each training fold.

The second dataset is a private enterprise SIEM log corpus collected from a production cloud environment over a 90-day window and anonymized before use. It contains 2.3 million log events, of which 47,812 are labeled as security-relevant by human analysts. The enterprise corpus was used for cross-source co-occurrence feature evaluation but is not publicly released for confidentiality reasons.

STIX threat intelligence was retrieved via TAXII 2.1 from MITRE ATT&CK v15 and from the OpenCTI community TAXII server. CVE and NVD records were fetched via the NVD REST API v2.0. Knowledge graph link prediction used GPT-4 (model version gpt-4-0125-preview) via the OpenAI API.

We ran three baselines. The first was a single-source CodeBERT classifier trained only on CVEfixes, which represents current best practice for LLM-based single-source detection. The second was XGBoost trained only on SIEM-derived features without any threat intelligence. The third was GPT-4 zero-shot classification on raw vulnerability descriptions with no knowledge graph layer at all. All experiments ran on an NVIDIA A100 80 GB GPU with 256 GB RAM and an AMD EPYC 7763 CPU, using five-fold stratified cross-validation. The complete code is in the companion repository [7].

V. RESULTS AND DISCUSSION

Table II shows classification results for all four systems under normal conditions and under FGSM perturbation at $\epsilon = 0.01$.

TABLE II: Classification Performance Comparison

System	Precision (%)	Recall (%)	F1 (%)	Adv. Acc. (%)
CodeBERT (single-source)	87.4	83.1	85.2	71.3
XGBoost (SIEM only)	82.6	79.8	81.2	68.9
GPT-4 zero-shot	89.1	84.7	86.8	N/A
CyberFuse-CI (proposed)	95.3	94.2	94.7	91.2

CyberFuse-CI outperforms all baselines by a meaningful margin. Against the strongest single-source baseline, GPT-4 zero-shot at 86.8%, the F1 gain is 7.9 percentage points. We attribute most of that gap to features that single-source classifiers simply cannot compute: cross-source co-occurrence counts, STIX temporal recency relative to CVE publication dates, and GAT and GPT-4 link confidence values from the knowledge graph.

The adversarial accuracy column is the part of Table II that matters most for production deployment. CodeBERT drops to 71.3% under FGSM perturbation and XGBoost SIEM-only falls to 68.9%. CyberFuse-CI holds at 91.2%, a gap of roughly 20 percentage points over the next best result. Adversarial training is the reason: the classifier is exposed to perturbed samples during training and does not internalize the gradient-aligned shortcuts that FGSM later tries to exploit.

Table III shows the zero-day candidate detection results. We define a zero-day candidate as any CVE entity that showed up in STIX threat actor reports before its NVD publication date.

TABLE III: Zero-Day Candidate Detection Metrics

Metric	CyberFuse-CI	STIX-Only	CVE-Only
Precision (%)	87.3	71.6	63.2
Recall (%)	84.1	68.3	59.7
Detection Lead Time (days)	6.2	2.1	0.0
False Positive Rate (%)	4.7	11.2	18.6

For us, the most striking result is the 6.2-day detection lead time. We're identifying vulnerabilities nearly a week before they receive official CVE numbers. That matters because it gives security teams a critical window to patch systems before exploit code goes public. Here's what we observed: threat intelligence indicators hit our TAXII feeds well before the NVD formally assigns CVE identifiers. The knowledge graph links those STIX indicator nodes to co-occurring SIEM anomaly nodes for the same affected component. A CVE-only tool sees none of that upstream signal. Our 4.7% false positive rate is also notably lower than STIX-only at 11.2% and CVE-only at 18.6%, which we attribute to the corroboration requirement across at least two source types before escalating a candidate.

Mean inference latency on the enterprise SIEM corpus was 11.4 ms per event. GPT-4 link validation is an offline batch step during graph construction and adds nothing to inference time. The online path uses only the pre-built graph and XGBoost.

We need to be candid about several limitations. First, the SIEM corpus comes from one organization on one cloud provider stack, and we haven't tested how well the normalization layer handles on-premises environments with different logging architectures. Second, the GPT-4 link validation step adds API cost that smaller security teams might find prohibitive. We estimate roughly \$0.15 per 1,000 vulnerability entities. Open-source alternatives like LLaMA-3 could reduce that cost, but we haven't validated their

performance yet. Third, our adversarial robustness testing used only FGSM perturbations. Stronger attacks like Projected Gradient Descent and Carlini-Wagner methods remain untested and represent important future work.

VI. GITHUB COMPANION REPOSITORY

The complete CyberFuse-CI implementation is publicly available in the companion repository [7]. The repository carries a v1.0.0 release tag corresponding exactly to the experimental configuration used in this paper.

The repository is organized into five production modules. The connectors module implements Layer 1 ingestion for all four data sources: the NVD REST API v2.0 connector, the STIX 2.1 and TAXII 2.1 bundle parser, the Syslog and CEF normalizer, and the CVEfixes SQLite loader. The knowledge_graph module implements the OWL ontology-aligned graph builder, the BiLSTM-CRF NER extractor with the 150-term cybersecurity domain dictionary, the Graph Attention Network link predictor, and the GPT-4 candidate link validator. The adversarial_training module provides the FGSM attack implementation, the adversarial training harness with configurable epsilon sweep, and the robustness evaluation reporter. The mitre_mapping module builds and queries the FAISS ATT&CK v15 embedding index. The evaluation module contains reproducible benchmark scripts against CVEfixes with five-fold cross-validation and CSV outputs matching Tables II and III.

VII. CONCLUSION

We built CyberFuse-CI on a simple premise: vulnerability signals scatter across data sources that don't communicate with each other. By fusing these streams through a knowledge graph and hardening the classifier against adversarial perturbations, we consistently outperform tools that look at single sources in isolation. The 94.7% F1-score and 91.2% adversarial accuracy show we don't have to choose between comprehensive coverage and robustness. We can achieve both simultaneously.

The 6.2-day zero-day detection lead time is, in our view, the result that translates most directly into organizational risk reduction. Deploying patches before public exploit code exists gives security teams a window that simply doesn't exist when working from CVE-only data. That lead time came from the knowledge graph connecting STIX indicators to SIEM anomalies on the same software component. We didn't tune for that metric specifically. It emerged from the architecture.

Our next steps are to test PGD and Carlini-Wagner adversarial attacks, run the SIEM connector against on-premises log formats, and compare open-source LLMs against GPT-4 for the knowledge graph link validation step.

VIII. ACKNOWLEDGMENTS

The authors thank the MITRE Corporation for maintaining the publicly available ATT&CK STIX repository and the CVEfixes research team for releasing their benchmark dataset.

AI-Generated Text Disclosure (per IEEE PSPB Policy): Portions of this manuscript, including drafts of Sections I through VII, were composed with the assistance of Claude, an AI language model developed by Anthropic [13]. All technical content, experimental claims, performance figures, and citations were independently verified and are the sole responsibility of the named authors. This disclosure is made in

accordance with the IEEE Publication Services and Products Board motion requiring disclosure of AI-generated text in IEEE Conference and Periodical submissions.

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